

Reduced or modified dietary fat for preventing cardiovascular disease.



BACKGROUND: Reduction or modification of dietary fat can improve total cholesterol levels, but may also have a variety of effects, both positive and negative, on other cardiovascular risk factors. OBJECTIVES: The aim of this systematic review was to assess the effect of reduction or modification of dietary fats on total and cardiovascular mortality and cardiovascular morbidity over at least 6 months, using all available randomized clinical trials. SEARCH STRATEGY: The Cochrane Library, MEDLINE, EMBASE, CAB s, CVRCT registry and related Cochrane Groups' trial registers were searched through spring 1998, SIGLE to January 1999. Trials known to experts in the field and biographies were included through May 1999. SELECTION CRITERIA: Trials fulfilled the following criteria: 1) randomized with appropriate control group, 2) intention to reduce or modify fat or cholesterol intake (excluding exclusively omega-3 fat interventions), 3) not multi factorial, 4) healthy adult humans, 5) intervention at least six months, 6) mortality or cardiovascular morbidity data available. Inclusion decisions were duplicated, disagreement resolved by discussion or a third party. DATA COLLECTION AND ANALYSIS: Rate data were extracted by two independent reviewers and meta-analysis performed using random effects methodology. Meta-regression and funnel plots were used. MAIN RESULTS: Twenty seven studies were included (40 intervention arms, 30,901 person-years). There was no significant effect on total mortality (rate ratio 0.98, 95% CI 0.86 to 1.12), a trend towards protection from cardiovascular mortality (rate ratio 0.91, 95% CI 0.77 to 1.07), and significant protection from cardiovascular events (rate ratio 0.84, 95% CI 0.72 to 0.99). The latter became non-significant on sensitivity analysis. Trials where participants were involved for more than 2 years showed significant reductions in the rate of cardiovascular events and a suggestion of protection from total mortality. The degree of protection from cardiovascular events appeared similar in high and low risk groups, but was statistically significant only in the former. REVIEWER'S CONCLUSIONS: The findings are suggestive of a small but potentially important reduction in cardiovascular risk in trials longer than two years. Lifestyle advice to all those at high risk of cardiovascular disease (especially where statins are unavailable or rationed), and to lower risk population groups, should continue to include permanent reduction of dietary saturated fat and partial replacement by unsaturates.